TermMap-DX: mapping extracted clinical terms to ICD-10-CM and SNOMED CT

HiLabs Hackathon 2026 · documentation and write-up

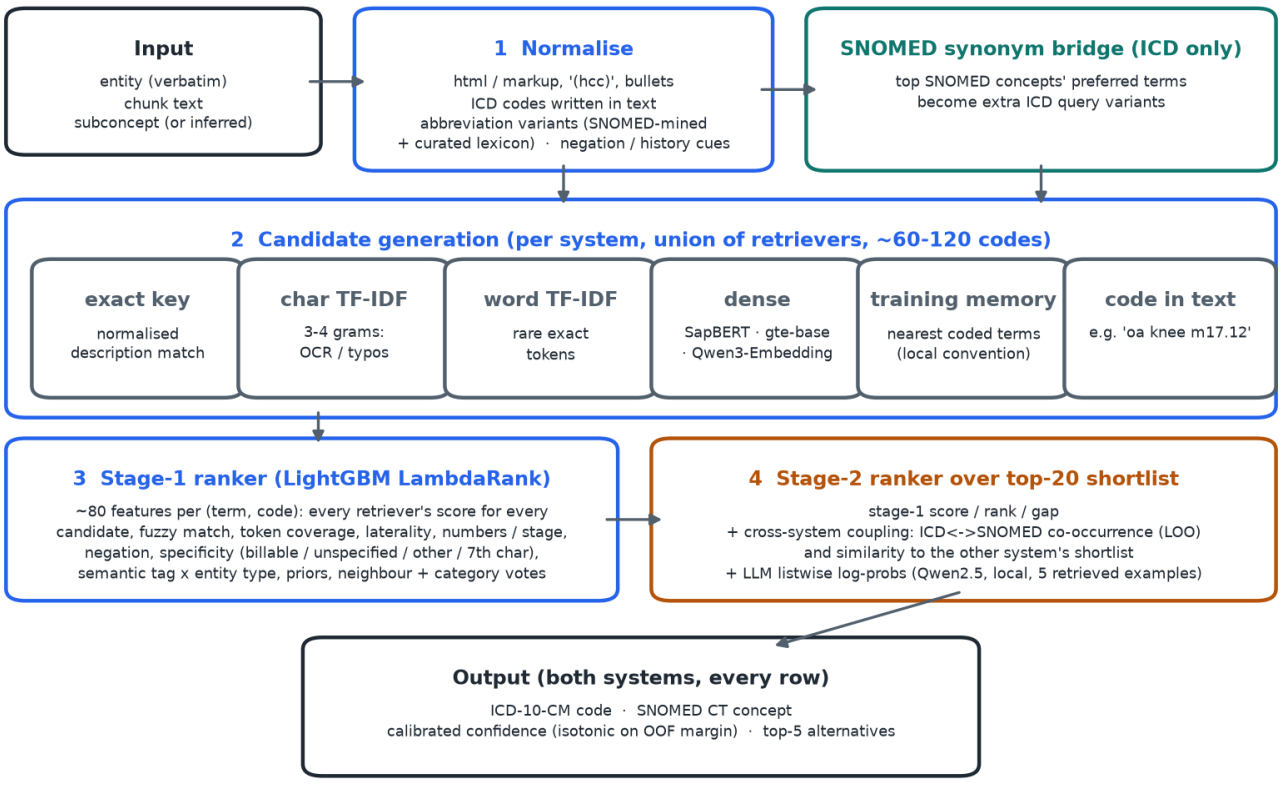
1. How we read the problem

The supplied training file differs from the schema in the brief, and that changes the problem:

- **Long format, one row per (entity, chunk, coding system).** 9,646 rows: 2,306 ICD-10-CM and 7,340 SNOMED CT. ICD rows are diagnoses only. SNOMED rows also cover procedures (2,777), labs (858), encounter/social items (567) and devices (437). 2,123 diagnosis terms are labelled in both systems.
- **Every entity string occurs exactly once per system.** The data is deduplicated by term, so the held-out set will consist of *unseen strings*. Memorising terms is worth nothing; the engine has to generalise.
- **The long tail is the bulk of the data.** 54% of gold ICD codes and 68% of gold SNOMED concepts occur exactly once. For 20% of ICD rows and 40% of SNOMED rows, the gold code appears nowhere else in training.
- **Gold labels encode a house convention.** Examples: "gait is slightly unstable" → R26.81, and "abnormal movement" → R29.898 (an "other symptoms" residual code). 98% of gold ICD codes are billable leaves; for SNOMED, 92% of gold `display` strings are the concept's preferred term. Some labels are arguably noisy ("benign prostatic hyperplasia" → *Benign fibroma of prostate*). Inter-annotator agreement is not published, so the practical ceiling is unknown.
- **Exact dictionary matching barely helps.** Only 4.3% of ICD and 18.9% of SNOMED diagnosis terms match a description of their gold code exactly after normalisation.

Scope: the engine maps **every** input row to one ICD-10-CM code *and* one SNOMED CT concept. Both come from the supplied dictionaries, which define the legal answer space. Every prediction carries a calibrated confidence and top-5 alternatives, so a downstream workflow can abstain or route low-confidence terms to review.

2. Architecture



Normalisation. Unescape HTML and strip markup (`<br>`), EHR artefacts (`(hcc)`, `cms-hcc`), bullets and list numbering. ICD codes written in the text are detected ("primary oa left knee m17.12") and become candidates. Abbreviation expansion produces *query variants*; it never replaces the original string. The lexicon has two sources: 7.7k `ABBR - Expansion` descriptions mined

from the supplied SNOMED file ("AR - Aortic regurgitation", "HHS - Hyperosmolar hyperglycaemic state") and a short curated list of chart shorthand (HTN, LE, bilat, s/p…). Ambiguous abbreviations (MS, RA, PID…) keep every reading as a separate variant, and the ranker uses context to choose. Negation, history and family-history cues are read from the 60 characters before the mention.

SNOMED synonym bridge for ICD. The SNOMED dictionary is roughly six times richer in synonyms than ICD. For ICD queries, the preferred terms of the top SNOMED concepts (dense match) are added as extra query variants. So "hepatic enlargement" also searches ICD as "hepatomegaly".

Candidate generation (recall). Per system, we take the union of: exact normalised-key match; character 3–4-gram TF-IDF (robust to OCR noise such as "chronickidney disease stag 3"); word TF-IDF (rewards rare exact tokens); dense retrieval with SapBERT, plus gte-base and Qwen3-Embedding in the GPU configuration; and the **nearest labelled training terms**, whose gold codes carry the house convention. All retrieval runs over unique normalised descriptions: 158k for ICD and 624k for SNOMED, after stripping the FSN semantic tag.

Stage-1 ranker. Every candidate is scored by *every* signal, not only by the retriever that proposed it. That gives ~80 features: max/preferred-term cosine per encoder (original and expanded query), char/word TF-IDF, fuzzy ratios, query/description token coverage, numbers and stage agreement ("stage 2" vs N18.30), laterality match/conflict/missing, negation and history agreement, and context overlap. For ICD it adds billable flag, code depth, "unspecified"/"other" wording, 7th-character encounter type and chapter. For SNOMED it adds semantic tag (crossed with the entity's subconcept) and description order. Training-memory features include neighbour votes at code and, for ICD, 3-character-category level (explicit specificity calibration), plus code and category priors. A LightGBM LambdaRank model per system ranks the candidates.

Stage-2 ranker. A second LambdaRank model works on the top-20 shortlist. It uses stage-1 score, rank and gap, plus **cross-system coupling**: the ICD and SNOMED predictions for the same term inform each other through (a) leave-one-out co-occurrence P(ICD | SNOMED) learned from the 2,123 doubly-labelled terms, and (b) dense similarity between each candidate and the other system's shortlist, weighted by its probabilities. In the GPU configuration, stage 2 also gets **LLM listwise scores**. A local Qwen2.5-Instruct model sees the term, its context, its entity type, the five most similar *already-coded* training terms with their codes, and the top-10 candidates as lettered options. One forward pass gives the log-probability of each option letter: no generation, no sampling, deterministic.

Calibration. Isotonic regression on the out-of-fold stage-2 margin turns ranker scores into P(correct), which is reported as `confidence`.

3. Evaluation protocol (no leakage)

All numbers are **5-fold cross-validation** on the training file. Folds are keyed by (entity, chunk, subconcept) and shared by both systems, so cross-system features never see the other system's label for the same term. Training-memory features (neighbour votes, priors, co-occurrence) are **leave-one-out**: a term is never scored against its own label. Stage-1 scores feeding stage 2 are out-of-fold. Early stopping uses an inner fold, never the evaluation fold. Exact-match accuracy is reported per system; the headline is the lower of the two.

RESULTS_SECTION

5. Error analysis

ERROR_SECTION

6. Design decisions and why

- **Learning-to-rank over hand-tuned score fusion.** No single signal wins. On SNOMED shortlists, SapBERT alone picks the gold concept 53% of the time, gte-base 52% and the lexical ranker 61%. The ranker learns *when* each signal is trustworthy (abbreviation, long phrase, procedure vs finding).
- **The training memory is a retriever, not a lookup.** Because every test string is new, labelled neighbours are used as *evidence* (similarity-weighted votes for a code and its category), so the convention ("other symptoms" residual codes, preferred SNOMED granularity) transfers to paraphrases.
- **Specificity as an explicit feature set.** "Unspecified"/"other" wording, billable leaf, code depth, 7th character, numeric and laterality agreement, and *category-level* neighbour votes let the ranker separate "right category, wrong leaf" from "wrong concept". It learns from data when the text supports a specific code and when E11.9-style unspecified codes are correct.
- **Both systems predicted independently, then coupled.** Each system has its own retrieval and ranker, so neither is a crosswalk of the other (the brief warns against that). Coupling enters only as stage-2 evidence.

- **LLM only where judgement is needed, and only as a feature.** The LLM never sees the whole vocabulary; it re-scores a 10-option shortlist in a single forward pass, and the ranker decides how much to trust it. Cost: about one 700-token forward pass per term per system.
- **Everything local.** The data is proprietary; no API calls are made, and all models are open-weight and named below.

7. What we tried and abandoned

- **Using the client's own index model as an oracle.** The SNOMED export notes say it came from a `gte_base` semantic index, so we tested whether gold labels follow gte-base's nearest neighbour. They do not (52% on shortlists vs 61% for our ranker). gte-base is kept as one feature, not as the decider.
- **BioLORD-2023** as an encoder. Dropped because its licence requires UMLS and SNOMED CT licences; replaced with Qwen3-Embedding-0.6B (Apache-2.0).
- **Stage 2 without new evidence.** Re-ranking stage-1 scores with only coupling features gives ±0.3 pt, which is within noise. Stage 2 earns its place only with the LLM scores.
- **Larger shortlists for stage 2.** Going from 12 to 20 candidates recovers 2–4 pt of recall without hurting top-1.
- **Official SNOMED→ICD map / UMLS.** Not used (licence-gated, and the brief notes the map cannot be ground truth). A data-driven co-occurrence prior replaces it.

8. Plan for the remaining gap

NEXT_SECTION

9. External resources

resource	version / id	use	licence
ICD-10-CM codes & descriptions	challenge file `icd10cm_codes.xlsx` (159,306 rows)	answer space, corpus	provided
SNOMED CT US codes & descriptions	challenge file (984,585 descriptions, 382,124 concepts; Term Map export)	answer space, corpus, abbreviation lexicon	provided
SapBERT	`cambridgeltl/SapBERT-from-PubMedBERT-fulltext`	dense retrieval + features	Apache-2.0
gte-base	`thenlper/gte-base`	dense features	MIT
Qwen3-Embedding	`Qwen/Qwen3-Embedding-0.6B`	dense features	Apache-2.0
Qwen2.5-Instruct	`Qwen/Qwen2.5-7B-Instruct`, `Qwen/Qwen2.5-14B-Instruct`	LLM listwise scoring	Apache-2.0
LightGBM 4.7, scikit-learn 1.9, rapidfuzz 3.14, sparse_dot_topn 1.2		ranking, retrieval	MIT/BSD

Not used: UMLS, the official SNOMED→ICD-10-CM map, any hosted API.